

AZURE DATA ENGINEERING PROJECT

**End-to-End Data Processing Pipeline using Azure
Data Factory and Databricks**

Based on

Olympics Games Analysis Dataset

Submitted By

Kanakanti Vidyyaa Akshaya

Course: Data Engineering

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1. INTRODUCTION

This project demonstrates the implementation of an automated Azure Data Engineering pipeline for processing Olympics datasets. The solution uses Azure Blob Storage, Azure Data Lake Storage Gen2, Azure Data Factory, Azure Databricks, and PySpark to automate data ingestion, storage, transformation, and processing.

1.1 SETUP CHECKLIST

Hardware:

- Windows System
- 4 GB RAM or above

Software:

- Microsoft Azure
- Azure Data Factory
- Azure Data Lake Storage Gen2
- Azure Databricks
- PySpark

1.2 INSTRUCTIONS

- Upload source CSV files into Azure Blob Storage.
- Configure Azure services.
- Create automated pipeline.
- Perform transformations using Databricks notebooks.
- Store processed files in Azure Data Lake Storage Gen2.

2. PROBLEM STATEMENT

2.1 OBJECTIVE

To build an automated data pipeline that ingests Olympics data from source CSV files, transfers it into Azure Data Lake Storage, performs PySpark transformations, and stores processed output files for further analytics.

2.2 ABSTRACT OF THE PROJECT

The project automates the movement and transformation of Olympics datasets using Azure cloud services. Data is collected from source CSV files, transferred through Azure Data Factory, processed using Databricks notebooks, and stored in Data Lake Storage for analytics and reporting.

2.3 TECHNOLOGIES USED

- Azure Blob Storage
- Azure Data Lake Storage Gen2
- Azure Data Factory
- Azure Databricks
- PySpark
- Storage Event Trigger

3. IMPLEMENTATION

3.1 SUMMARY OF FUNCTIONALITY

The system automatically detects newly uploaded Olympics files and executes an end-to-end pipeline for data ingestion and transformation.

3.2 PROJECT FLOW

Source CSV Files



Azure Blob Storage



Azure Data Factory



Azure Data Lake Storage Gen2



Azure Databricks Notebook



PySpark Transformations

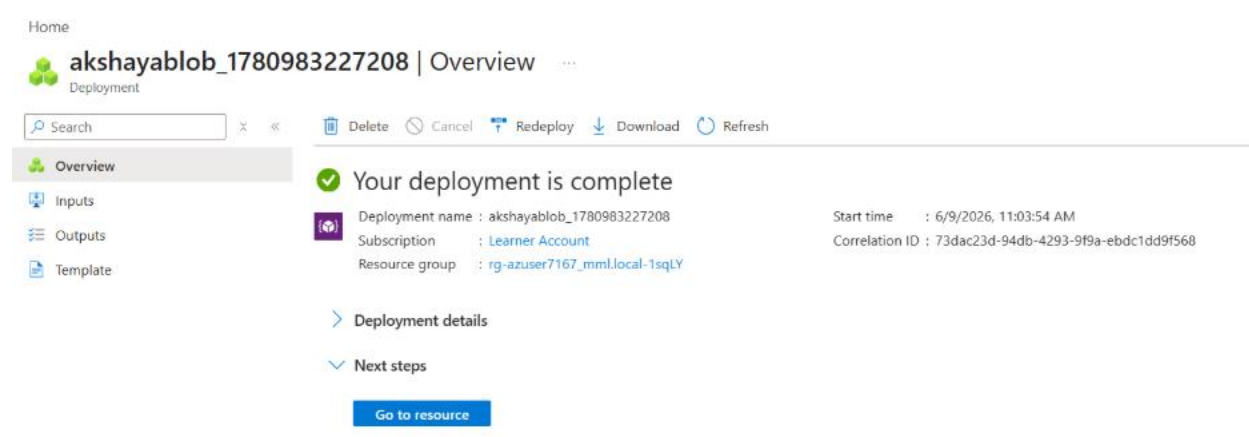


Processed Output Files

3.3 STORAGE ACCOUNT CREATION

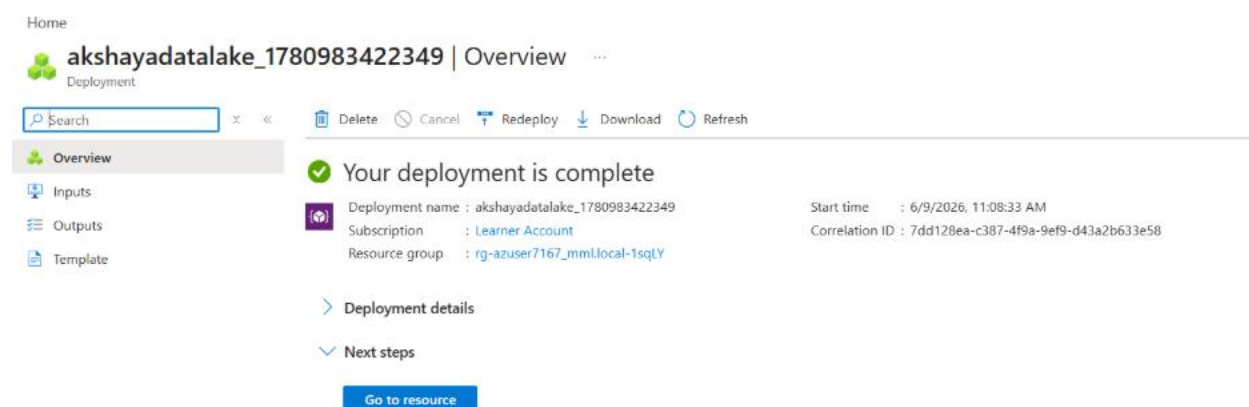
Azure Blob Storage

Azure Blob Storage was configured as the source storage layer for the project. All input CSV files were uploaded to the **Source** container and served as the starting point for the automated data ingestion pipeline.



Azure Data Lake Storage Gen2 (ADLS Gen2)

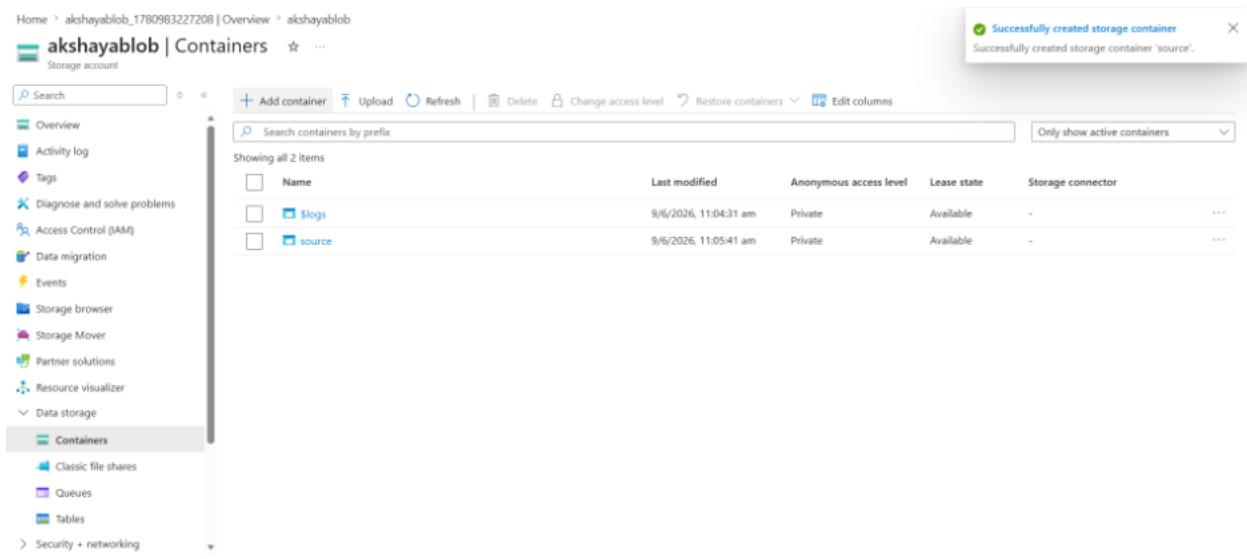
ADLS Gen2 was used as the destination storage system for the project. Files transferred through Azure Data Factory and processed datasets generated through PySpark transformations were stored in the **Destination** container for further analysis and reporting.



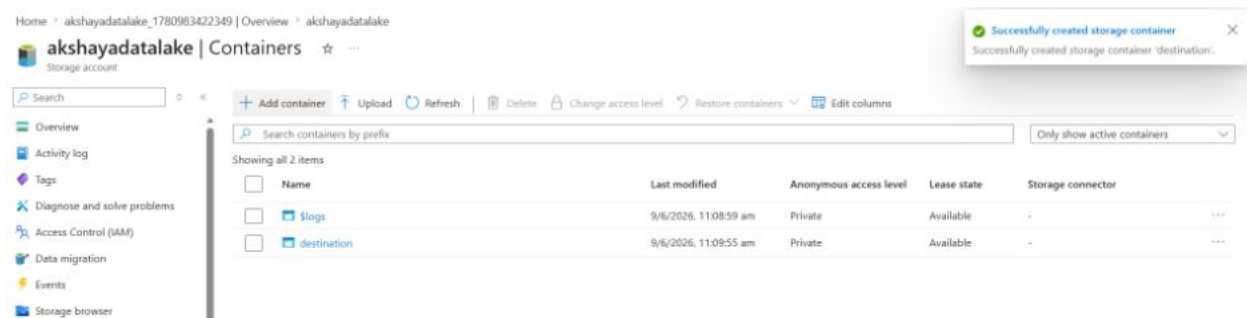
3.4 CONTAINER CREATION

Created:

- source container



- destination container

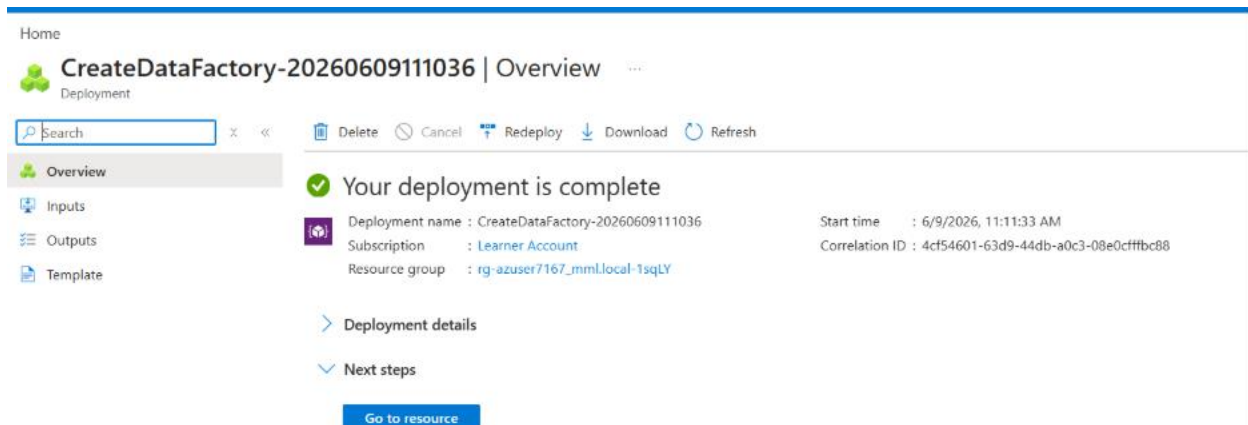


Source container stores incoming CSV files.

Destination container stores copied and transformed files.

3.5 AZURE DATA FACTORY CONFIGURATION

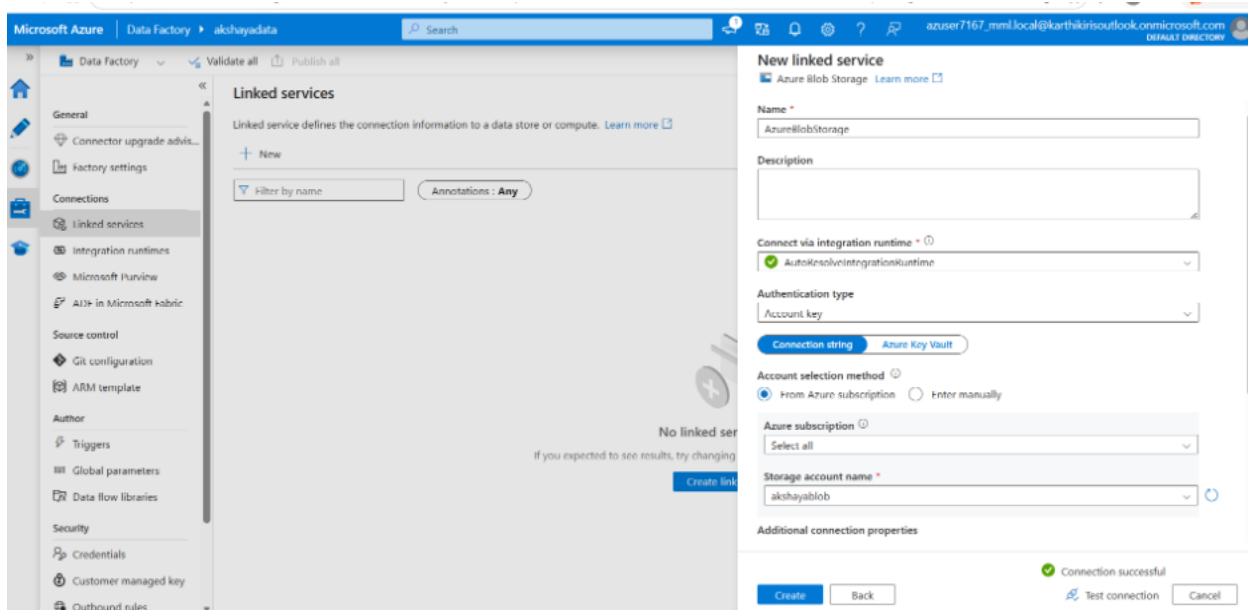
Created Azure Data Factory instance and configured Integration Runtime.



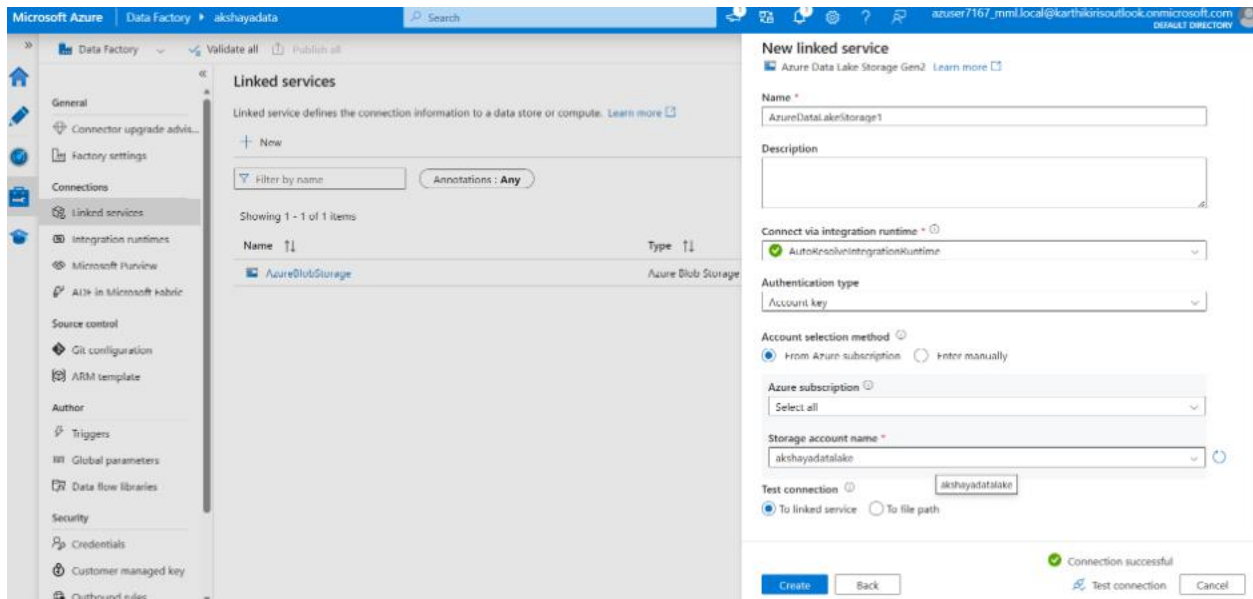
3.6 LINKED SERVICES CREATION

Configured linked services for:

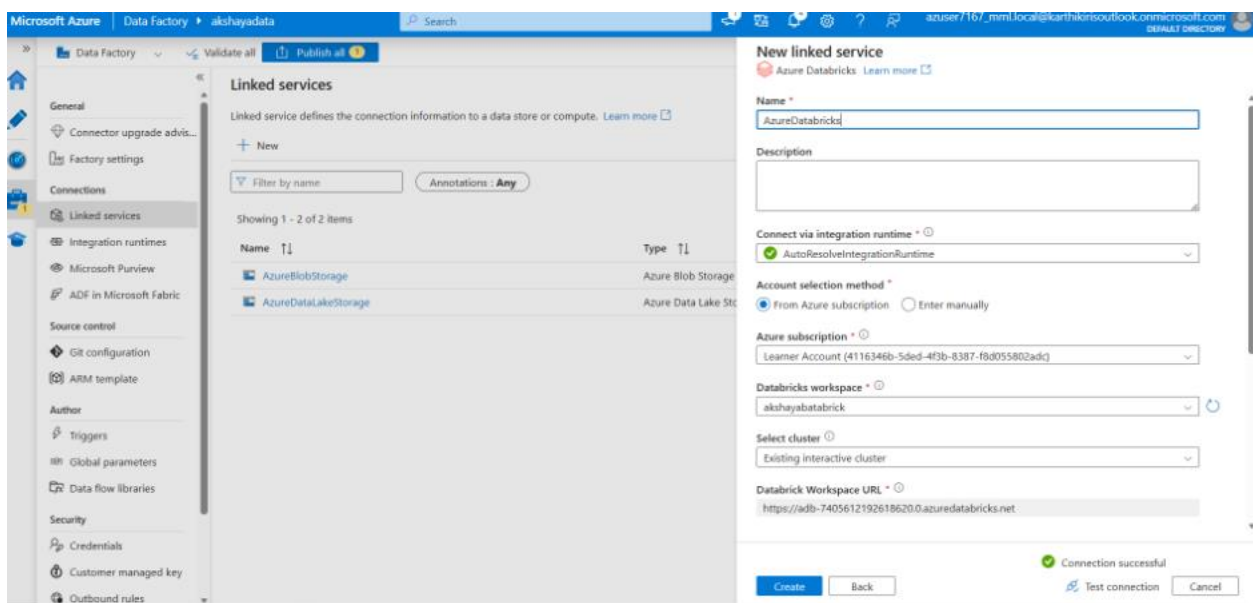
- Azure Blob Storage



- Azure Data Lake Storage Gen2



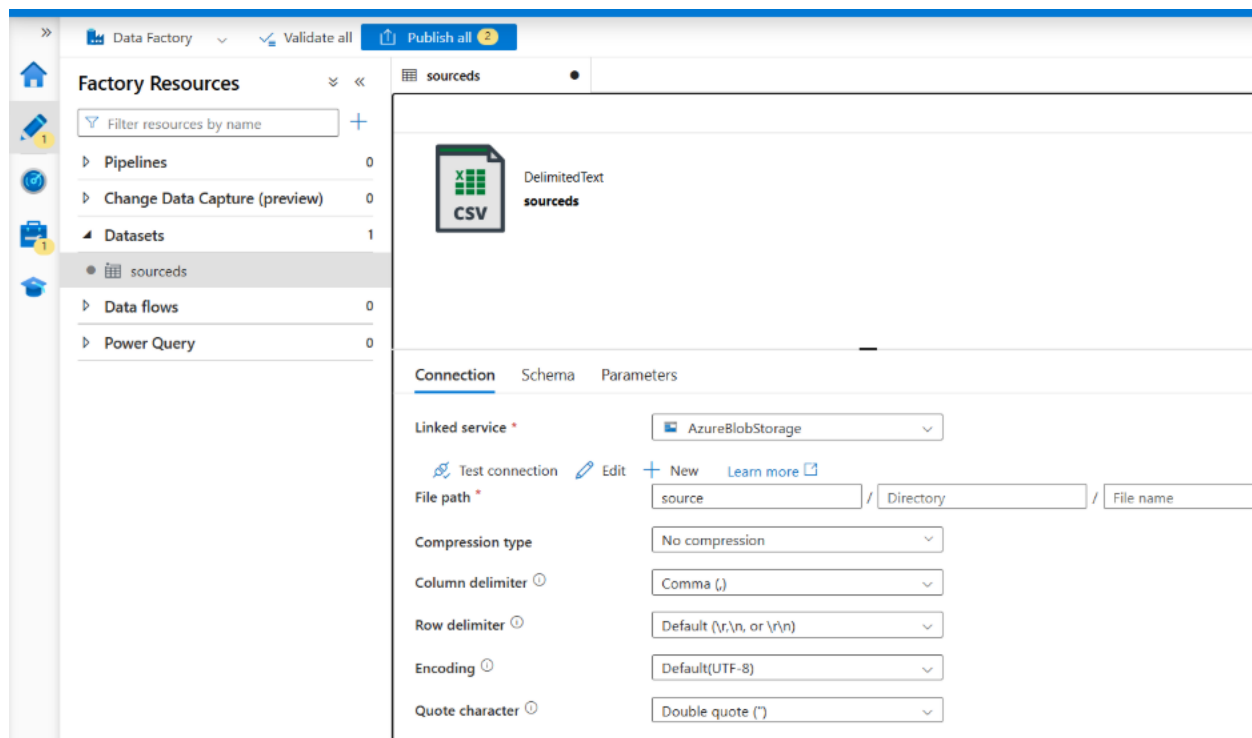
• Azure Databricks



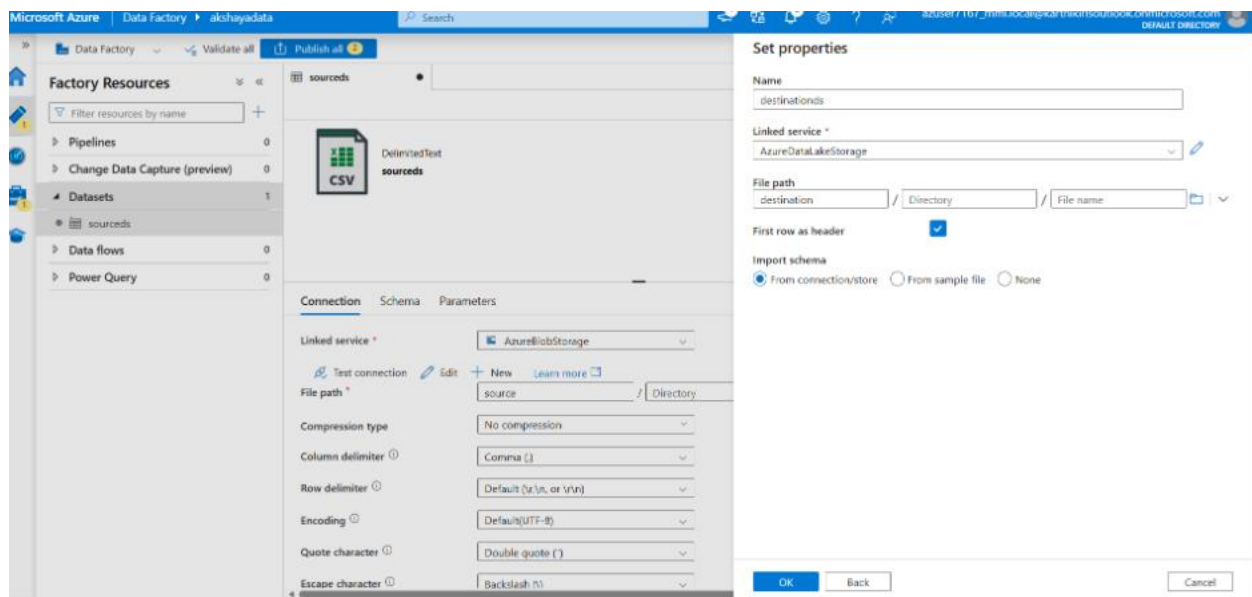
DATASET CREATION

Created datasets:

- sourced



- destinationds

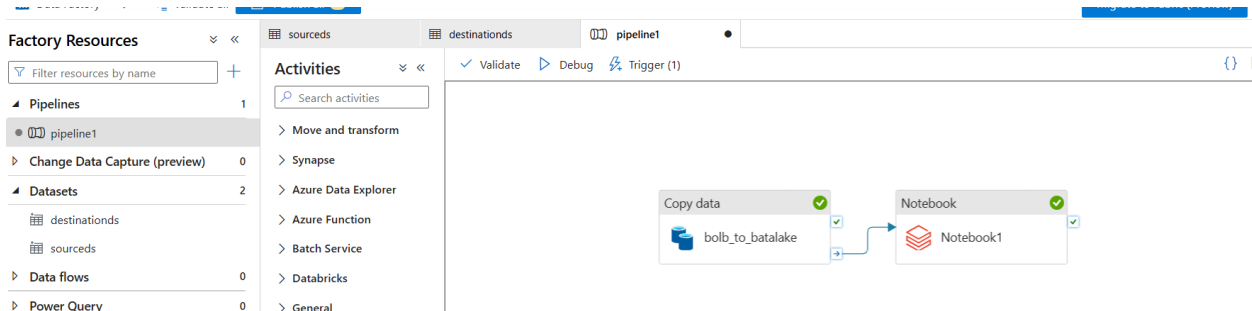


Both datasets use CSV format.

PIPELINE DEVELOPMENT

Created pipeline containing:

- Copy Data Activity
- Databricks Notebook Activity



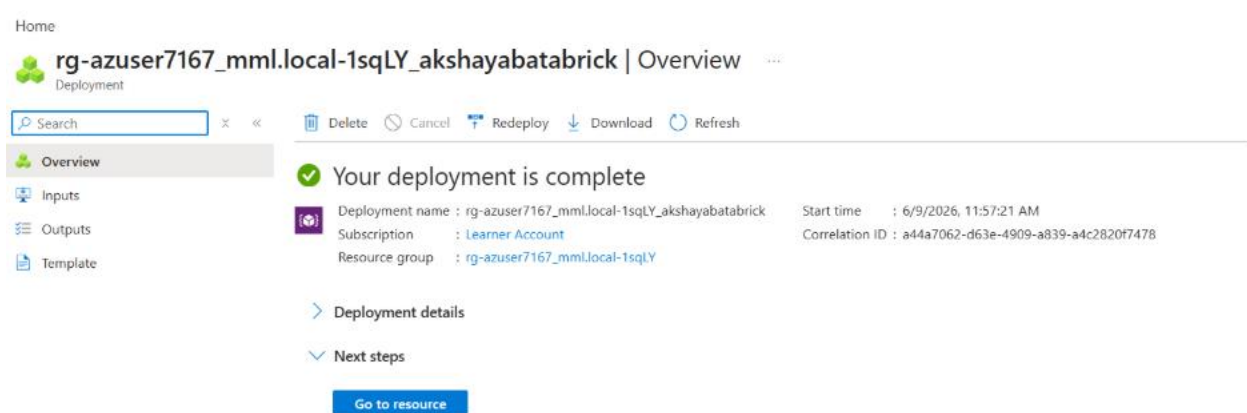
The Copy Activity transfers files from Blob Storage to Data Lake.

The Notebook Activity executes PySpark transformations.

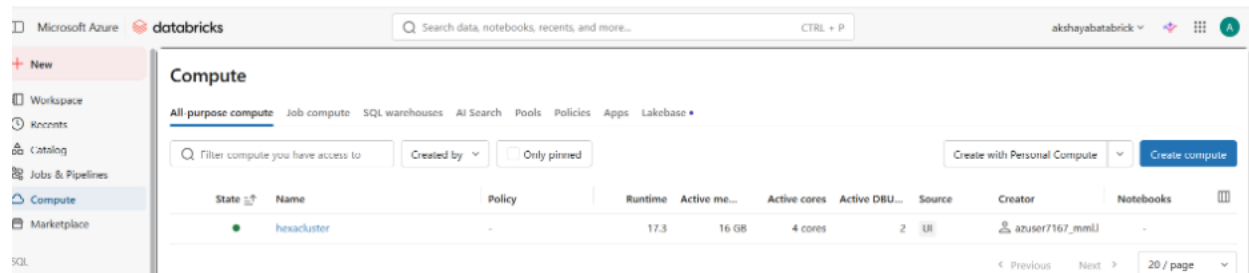
3.9 DATABRICKS CONFIGURATION

Created:

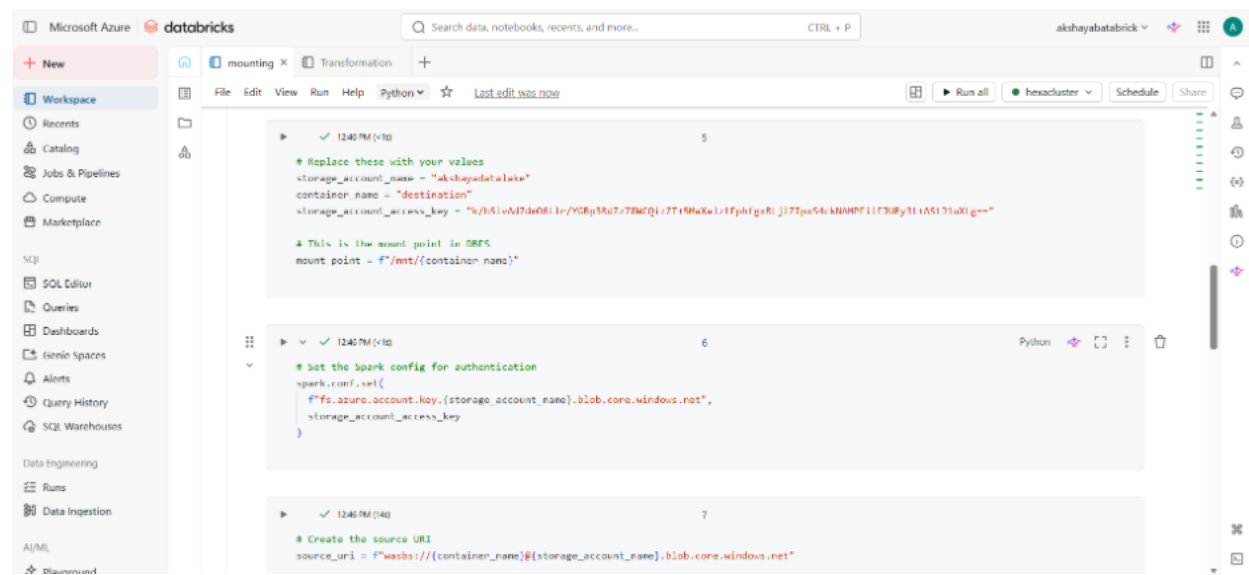
- Databricks Workspace



- Compute Cluster



- Mounting Notebook



- Transformation Notebook

Microsoft Azure

databricks

Search data, notebooks, recents, and more...

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

AI/ML

Playground

Agents

Runs >

ADF_akshayadata_pipeline1_Notebook1_e38eb546-bf45-426c-a7c5-b4c716

Output

Hide code

Export as HTML

Clone into new notebook

✓ 1.88s 4

country_medals.write \

.mode("overwrite") \

.option("header", "true") \

.csv("/mnt/processed/country_medal_summary")

▶ (2) Spark Jobs

✓ 6.39s 5

country_medals.write \

.format("delta") \

.mode("overwrite") \

.save("/mnt/processed/delta_country_medals")

▶ (2) Spark Jobs

✓ 3.64s 6

spark.sql("""

CREATE TABLE IF NOT EXISTS olympics_country_medals

USING DELTA

LOCATION '/mnt/processed/delta_country_medals'

3.10 DATA TRANSFORMATION

The following transformations were performed:

- Read source CSV files
- Validate schema
- Remove duplicates
- Filter unwanted records
- Select required columns
- Standardize column names
- Apply business transformations
- Store processed output files

3.11 TRIGGER CONFIGURATION

Created Storage Event Trigger.

Whenever a new file is uploaded to the source container, the pipeline executes automatically.

The image displays two screenshots from the Azure portal. The top screenshot shows the 'New trigger' configuration in Azure Data Factory. The trigger is named 'trigger', has a description, and is of type 'Storage events'. It is configured to use the 'From Azure subscription' account selection method. The Azure subscription is 'Ismar Account [4116544b-5d6d-4f6b-8687-f1ed05a8b7ad6]'. The storage account name is 'akshayablob', and the container name is '/source/'. The blob path begins with '/source/'. The trigger is set to be 'Activated' with a timeout of '0.12:00:00' and a retry count of '0'. The bottom screenshot shows the 'destination' container in Azure Storage Explorer. It displays a table of blobs with columns: Name, Last modified, Access tier, Blob type, Size, and Lease state. The table shows two items: 'country_medal_summary' and 'delta_country_medals', both created on 9/6/2026.

Home > akshaydatalake | Containers

destination ...

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

destination

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 4 items

Name	Last modified	Access tier	Blob type	Size	Lease state
processed	9/6/2026, 7:13:09 pm				

Home > akshaydatalake | Containers

destination ...

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

destination > processed

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					
country_medal_summary	9/6/2026, 7:14:00 pm				
delta_country_medals	9/6/2026, 7:21:56 pm				

4. RESULTS

- Source files successfully uploaded to Azure Blob Storage.
- Files copied successfully to Azure Data Lake Storage Gen2.

- Databricks notebook executed successfully.
- PySpark transformations completed successfully.
- Processed output files stored successfully.
- Automated trigger execution verified successfully.

4.1 CONCLUSION

This project successfully implemented an automated Azure Data Engineering pipeline using Azure Blob Storage, Azure Data Lake Storage Gen2, Azure Data Factory, Azure Databricks, and PySpark. The Olympics dataset was ingested, transformed, and stored successfully, while the Storage Event Trigger enabled automatic pipeline execution. The project provided hands-on experience in building a complete cloud-based data processing workflow.

5. APPENDIX

Source File

- **OlympicsAnalysis.csv**
- Source file used for data ingestion and transformation.

Output Files

- **country_medal_summary.csv**
- Contains country-wise medal summary generated after data processing.
- **delta_country_medals**
- Delta table created from the transformed Olympics data.

Azure Services Used

- Azure Blob Storage
- Azure Data Lake Storage Gen2
- Azure Data Factory
- Azure Databricks
- PySpark

Project Outcome

- Successfully ingested source data.

- Successfully copied files from Blob Storage to Data Lake Storage.
- Successfully transformed data using PySpark.
- Successfully generated CSV and Delta outputs.
- Successfully automated the pipeline using a Storage Event Trigger.